

Asher Siddiqui

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EDUCATION

FAST - NUCES Bachelor of Software Engineering (6th semester)

Expected Graduation: 2025

Key Courses: Computer Networks, Database Systems and & Design, Design & Analysis of Algorithms, Operating Systems, Data Structures, Web Engineering, Object Oriented Programming, Software Design & Architecture, Software Quality Assurance

SKILLS

Technical: Go Lang, ExpressJS, NodeJS, REST APIs, Oracle Database, MySQL, Git, Bash, Linux, Docker, Spring Boot, Java

Soft: Organization, Time Management, Communication, Persistence, Problem Solving, Willingness to learn

WORK EXPERIENCE

FAST - NUCES

Jan 2024 - Present

Teaching Assistant - Operating Systems

- Demonstrated strong organization, communication, and time management skills by efficiently working with instructors to manage the grades of 90 students.
- Programmed an assignment evaluation tool, producing significant time savings and highlighting problem solving skills.

PROJECTS

github.com/pluuto19

Load Balancer [Go Lang, ExpressJS, Docker, Git, HTML]

- Designed and implemented a high-throughput HTTP reverse proxy in Go, achieving efficient traffic distribution as demonstrated by the integration of round-robin and least connection algorithms.
- Enhanced application scalability and reduced computation overhead, as demonstrated by a throughput of 10000 requests/5 seconds, by utilizing goroutines and buffered channels in Go.
- Boosted load balancer's reliability by successfully implementing health checks to ensure backend servers' availability.

Project Management System [ExpressJS, MySQL, Handlebars, Git]

- Developed a robust webserver architecture for a project management system using ExpressJS and MySQL, utilizing the microservice architecture with three distinct services.
- Facilitated seamless interaction and data flow within the application by orchestrating communication between services through the integration of the controller with backend APIs.
- Ensured secure access across the system's services by employing API keys and JWT authentication with the REST APIs and controller, respectively.

Visual Convex Hulls [Java, JavaFX, Git]

- Demonstrated practical skills by translating research papers into applications using JavaFX.
- Conducted an analysis of computational complexity by implementing and executing various convex hull algorithms.

Miscellaneous [Python, Bash, ExpressJS, SQL, Java, Java Swing, Git, HTML]

- **Pyonic Reader:** A Python CLI tool for parsing PDFs, bolding initial letters of every word to help an ADHDer's reading focus.
- **Assess Buddy:** A bash tool for streamlined extraction, compilation, and execution of compressed folders, facilitating evaluation of student submitted source code.
- **Rent a Car & Autoshop:** An Express-based app with a normalized MySQL database for efficient CRUD operations.
- **A Night in the Woods:** A 2D platformer game made entirely in Java Swing, multithreaded to support background audio.

LEADERSHIP EXPERIENCE

Microsoft Learn Students Ambassador

Oct 2023 - Present

Technical Coordinator

- Coordinated technical activities for a workshop with approximately 50 participants. curating and managing workshop materials for the audience.
- Actively assisted the event management team during the workshop, ensuring smooth execution of the workshop.

VOLUNTEER EXPERIENCE

Team Guide : Mentored K-12 students at Destination Imagination Pakistan 2024, fostering creativity and problem-solving skills, while facilitating campus navigation. Contributed to a positive team experience, resulting in strengthened leadership and mentorship abilities.

AWARDS & HONORS

Dean's List of Honors - Fall 2023